.NET Migration Guide

GstarCAD 2020



GstarCAD

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1. Brief Introduction

Secondary developers from various industries have developed mass of .NET applications base on AutoCAD®, and one of the highlights of GstarCAD .NET is that GstarCAD achieves source-level compatibility with AutoCAD® .Net programs. Which means a set of code suits for two platforms, the developers can migrate the .NET program from AutoCAD to GstarCAD with almost no code changing. This guide provided for developers migrating their code to GstarCAD®.

2. Development Environment

- Microsoft Visual Studio Enterprise 2017（Version 15.9.17）
- CPU：

Basic Requirement：1.6 GHz CUP，Recommend：3.0 GHz CPU and above

- RAM：

Basic Requirement：2 GB，Recommend：8 GB and above

- Operation System

Windows 10 version 1507 and advanced version：Including home version, professional version, education version and enterprise version（not support LTSC and Windows 10 S）

Windows 8.1（with updated 2919355）： Core version, professional version and enterprise version

Windows 7 SP1（with latest Windows updated）： Including home version, professional version, enterprise version and Ultimate version

- Monitor Resolution:

Traditional monitor：1028 x 800 and above CRT monitor, High resolution and 4K (3840 x 2160) monitor

- GRX SDK 2020
- GstarCAD 2020
- .NET Framework 4.8 and above

3. Original Code

The original code should be DOTNET application source code which is from the DONET ObjectARX 2010 or higher version.

4. Install GstarCAD SDK

Download GstarCAD SDK package from our official website:

https://www.gstarcad.net/download/

After you get the GRXSDK.ZIP file, please extract it to your disk directory, for instance, C:\grxsdk. Afterwards you will get 6 files, which they are inc,inc-x64,inc-x86,lib-x64,lib-x86 and samples.

- The Inc file includes header files
- The inc-x64 file includes the files used in COM and .NET for 64 bit
- The inc-x86 file includes the files used in COM and .NET for 32 bit
- The lib-x64 file includes the files referenced from GRX library
- The lib-x86 file includes the files referenced from GRX library
- Samples includes project files, they are dotnet，fact_dg，HelloADS，HelloARX，SimplePalette, etc.

The instruction of sample projects:

- Dotnet includes multiple dotnet samples，including addline，hello，vbhello, etc.
  1) Addline includes .NET example for adding solid line.
  2) Hello includes .NET example for prompting the information of output.
  3) Vbhello includes VB.NET example for how to develop with .NET API.
- fact_dg includes example for defining LSP function.
- HelloADS includes the project for how to develop the type of ADS.
- HelloARX includes includes the example for how to develop with ARX API.

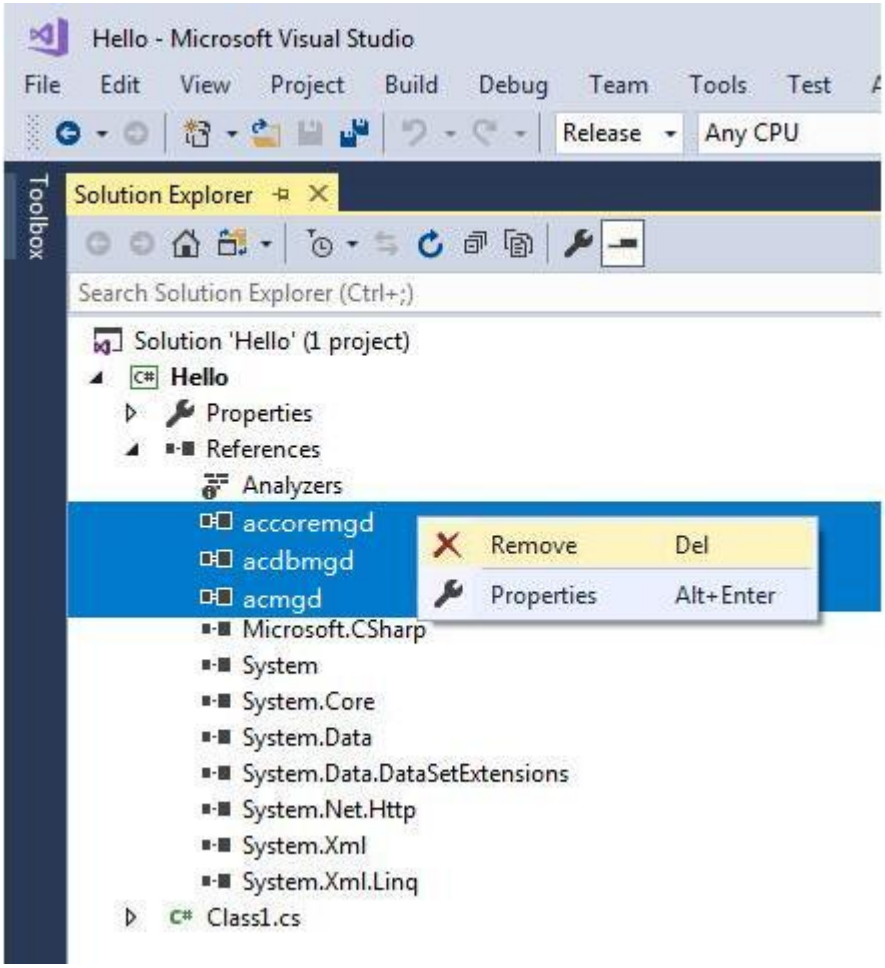
SimplePalette includes the example for how to develop the type modules for palette

5. VS2017 C# .NET Migration Process

5.1. Change Reference Files

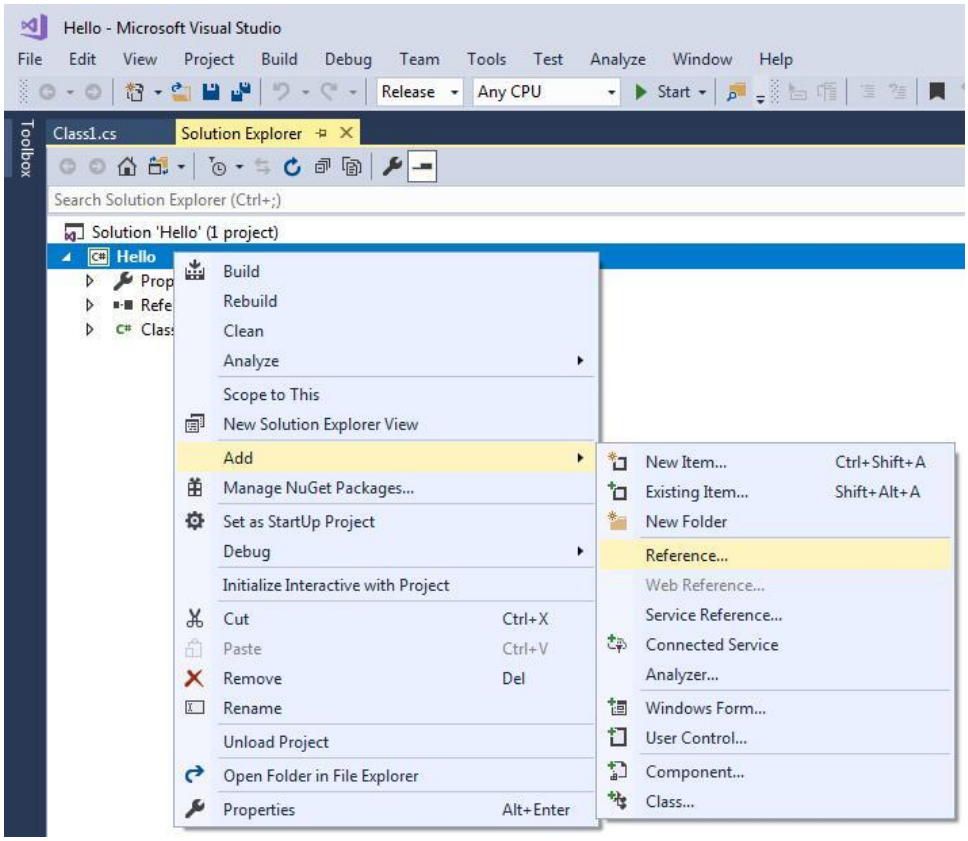
5.1.1. Remove the Reference

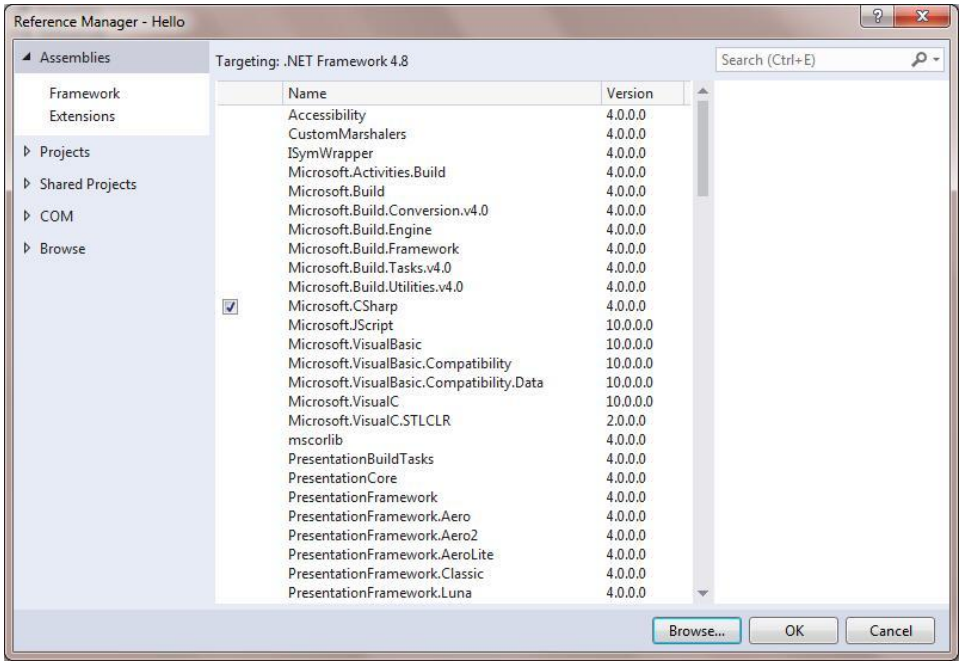
Remove acad reference files, as shown below.



5.1.2. Add Reference

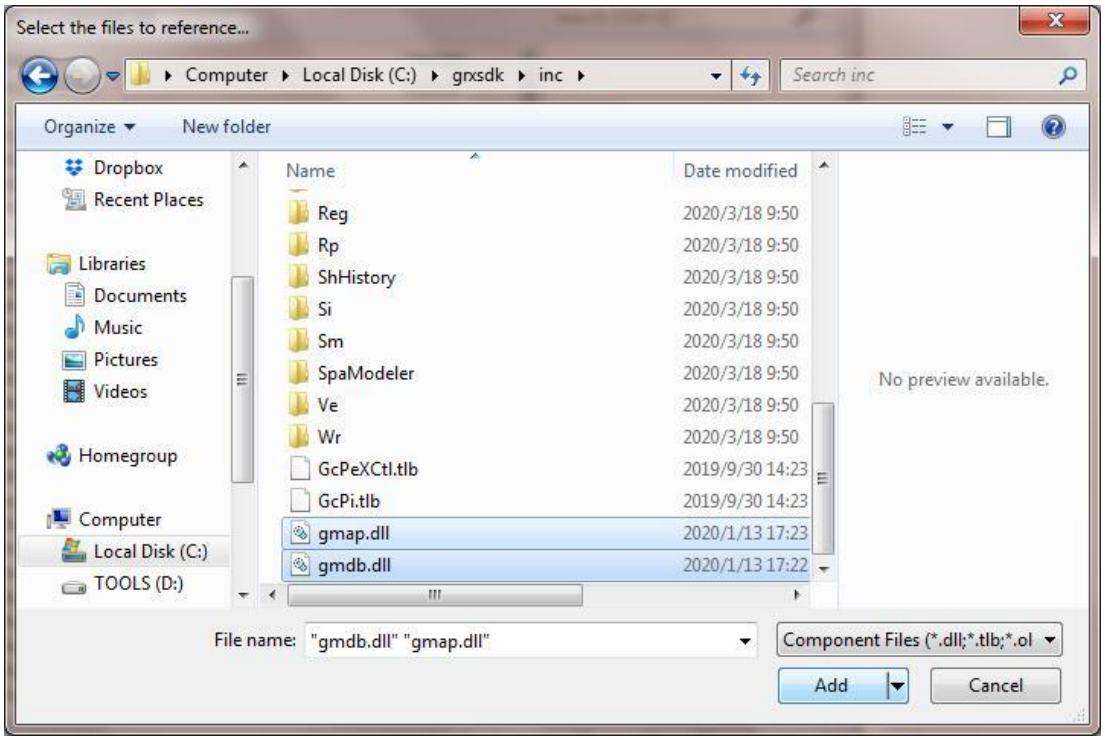
In VS 2017, find the Hello project you have been created on the solution explorer. Select the Hello project and right click to select Add Reference. After the "Reference Manager" dialog box pops out, click Browse button as shown below.

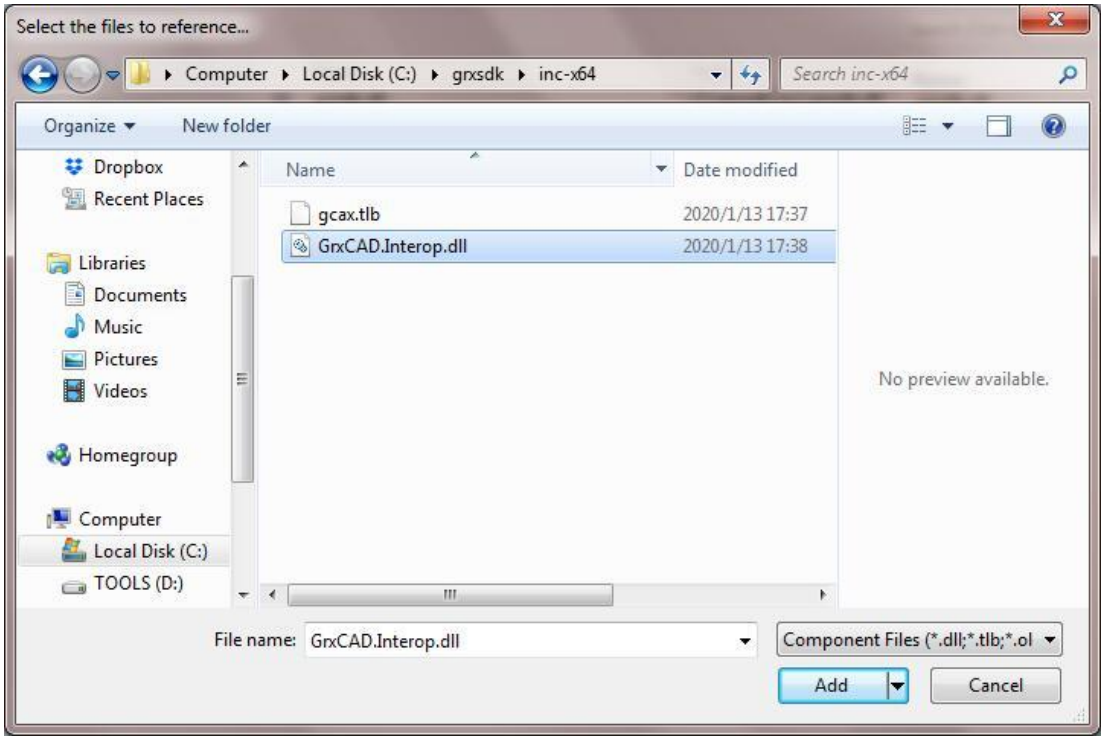




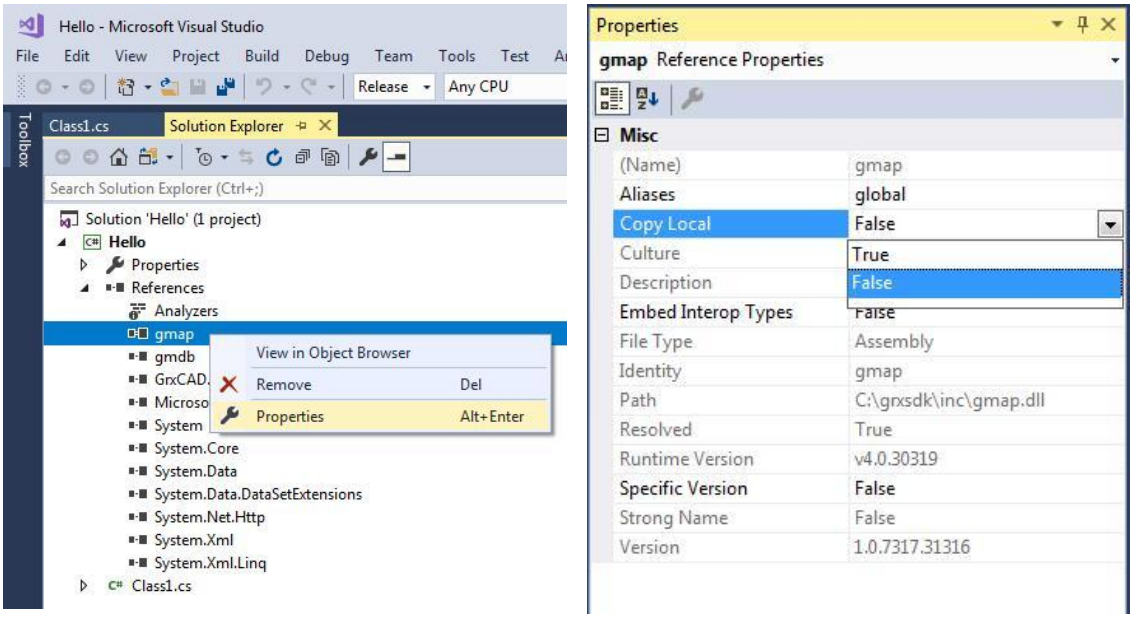
5.1.3. Choose and Add the Files

Find the GstarCADSDK installation path, for example, “C:\grxsdk\inc”. Select gmap.dll and gmdb.dll files in “c:\grxsdk\inc-x86” and add them by clicking Add button（If you use GstarCAD 64-bit version, please select GrxCAD.Interop.dll in inc-x64）



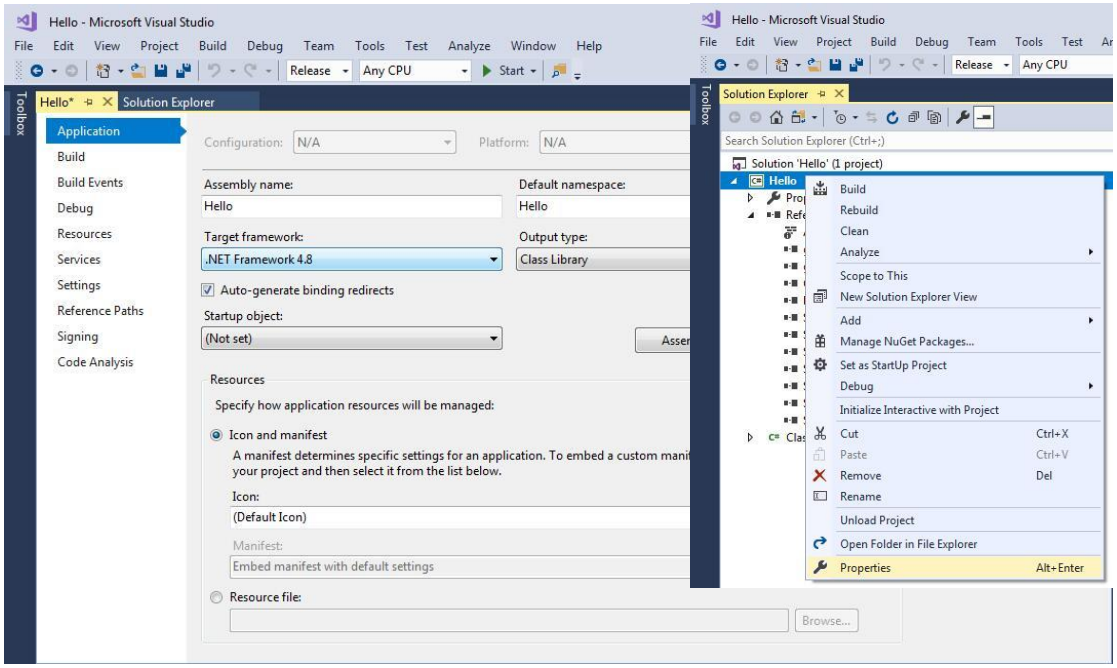


Remark：gmap.dll and gmdb.dll have to be added but file GrxCAD.Interop.dll is optional. After adding the files, right click each added reference file, click Property at the context menu, then the "Reference Properties" palette pops up, change the "Copy Local" property from True to False. The following image shows how to change "Copy Local" property for gmap.dll file. Please take it as reference to change gmdb.dll file's Copy Local property.



5.1.4. Change the Target Framework

In VS 2017, find the Hello project you have been created on the solution explorer. Right click to select Properties. After the new page opens, please change the "Target framework" to Framework 4.8 from the pull-down list in Application menu.



5.1.5. Modify the Name in Name Space

Replace the Autodesk.AutoCAD corresponding Name sapce to GrxCAD.

5.1.6. Compile Program

Click Visual Studio 2017 menu item [Build] » [Build solution], then you will get a “Hello.dll” file at the corresponding directory.

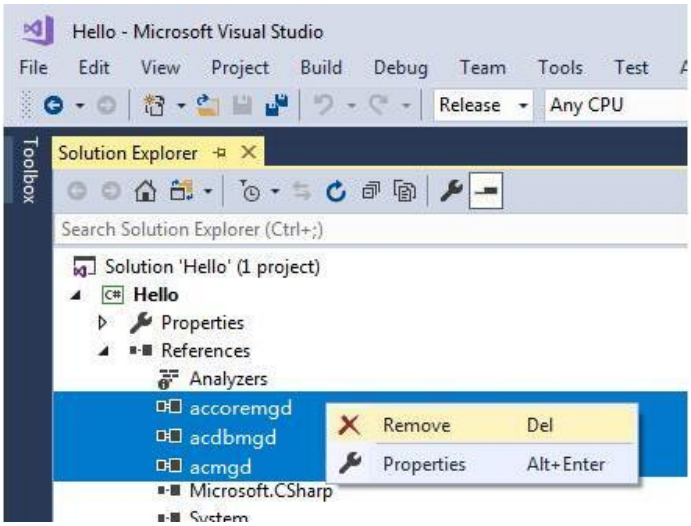
Run GstarCAD and input "netload" at command line, there will a "Select .Net Assembly" dialog box pops out, select the new generated ".dll" file and click Open button to load it.

6. VS2017 VB.NET Migration Process

6.1. Change Reference Files

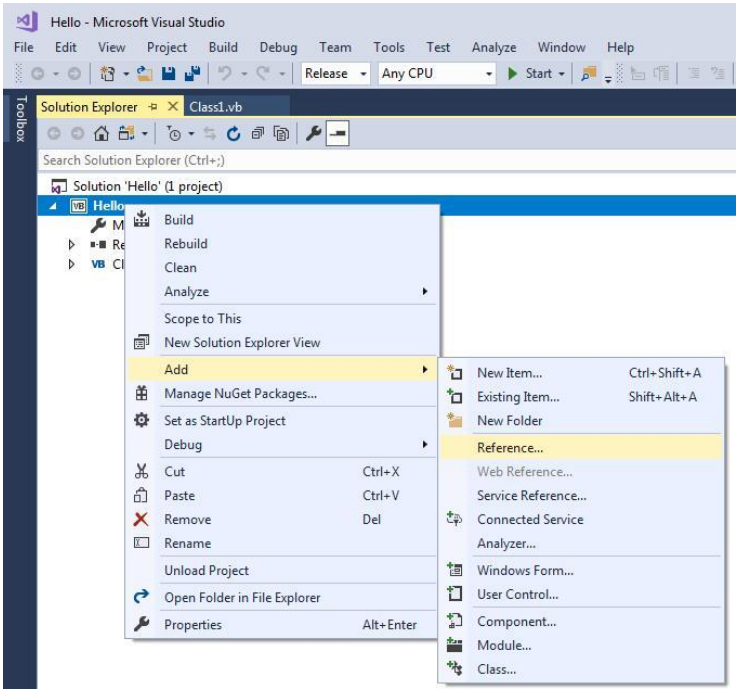
6.1.1. Remove the Reference

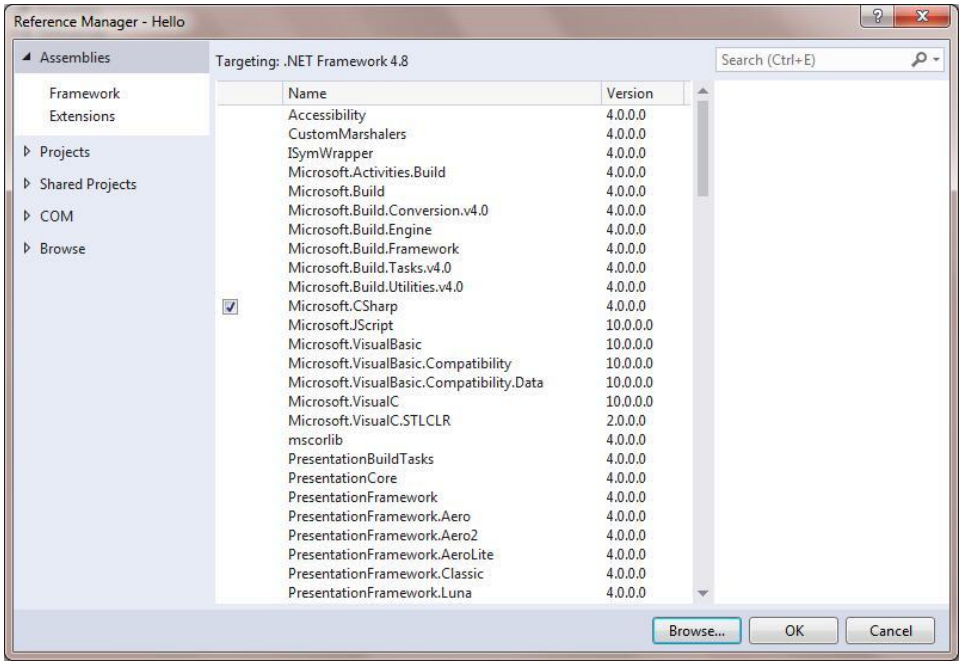
Remove acad reference files, as shown below.



6.1.2. Add Reference

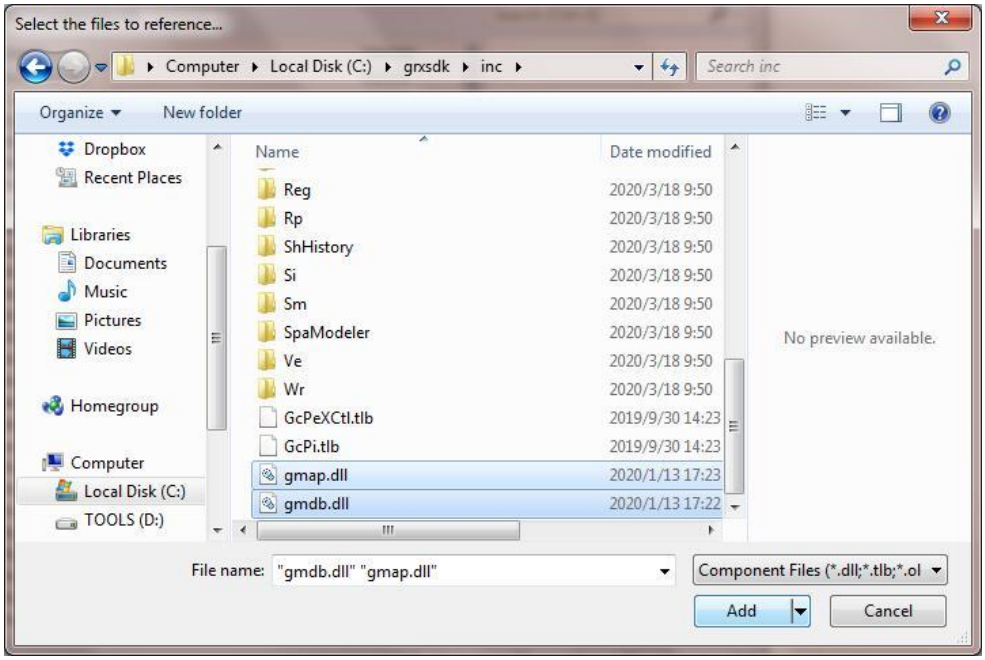
In VS 2017, find the Hello project you have been created on the solution explorer. Select the Hello project and right click to select Add Reference. After the "Reference Manager" dialog box pops out, click Browse button as shown below.





6.1.3. Choose and Add the Files

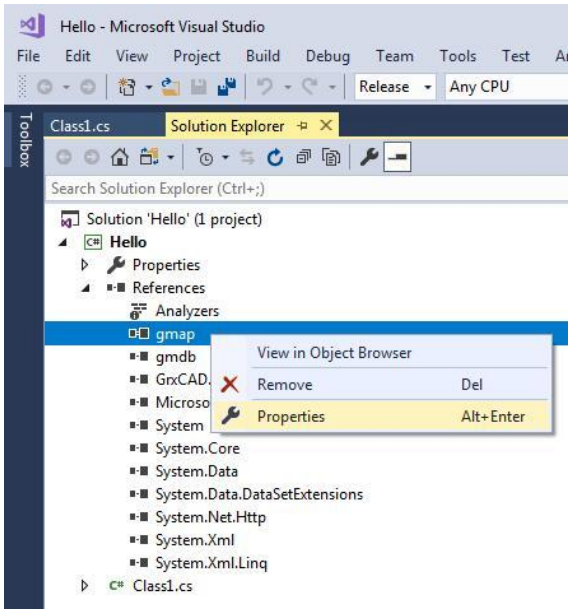
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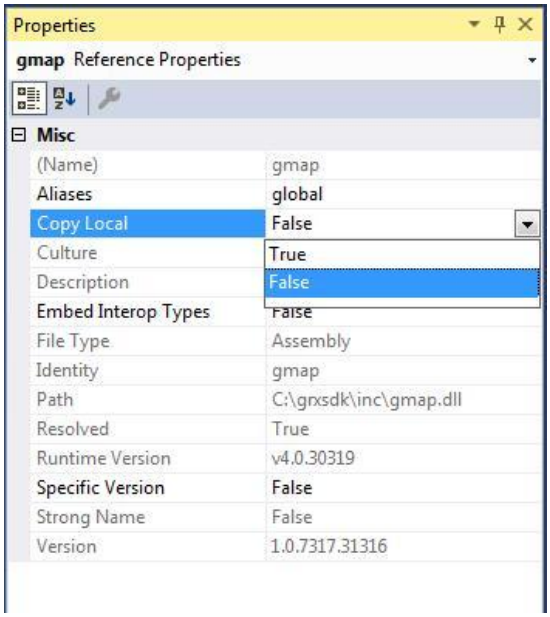




Remark：gmap.dll and gmdb.dll are necessary，GrxCAD.Interop.dll is alternative.

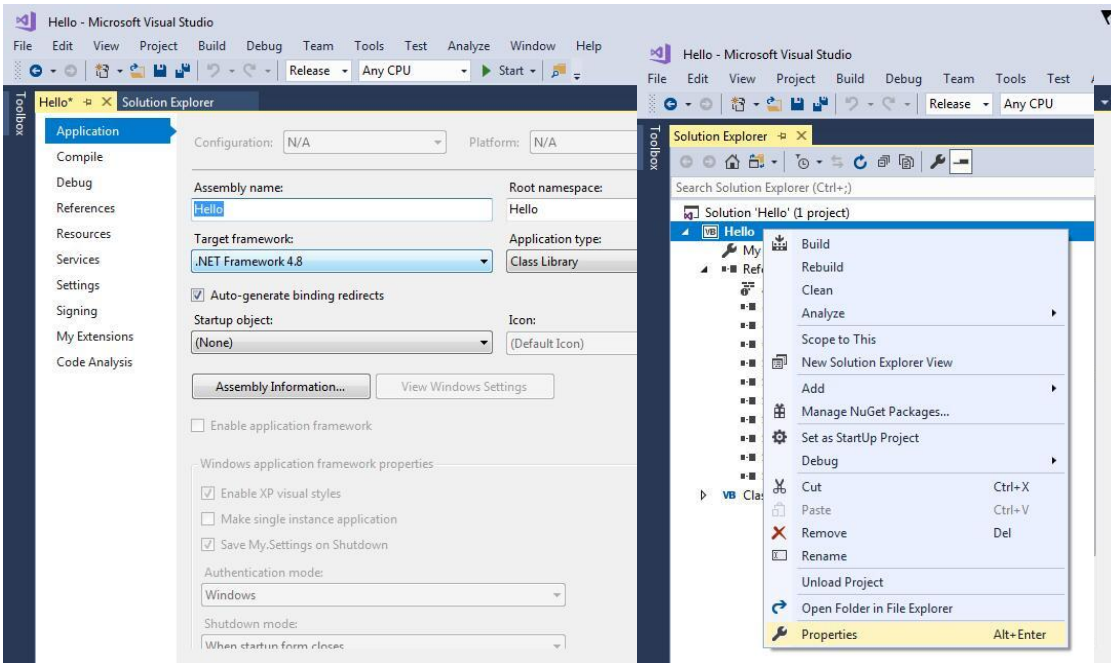
After adding the files, right click each added reference file, click Property at the context menu, then the "Reference Properties" palette pops up, change the "Copy Local" property from True to False. The following image shows how to change "Copy Local" property for gmap.dll file. Please take it as reference to change gmdb.dll file's Copy Local property.





6.1.4. Change the Target Frame

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6.1.6. Compile Program

Click Visual Studio 2017 menu item [Build] » [Build solution], then you will get a “Hello.dll” file at the corresponding directory.
Run GstarCAD and input "netload" at command line, there will a "Select .Net Assembly" dialog box pops out, select the new generated ".dll" file and click Open button to load it.

6.2.Using

Code like “var mapping = new IdMapping()” need to add “using” function first.

For example “var mapping = new IdMapping()” in GstarCAD should be：
using(var mapping = new IdMapping())

{

...

}

6.3. ResultBuffer

Use **ResultBuffer.ResbufObject** to replace **ResultBuffer.UnmanagedObject**;

ResultBuffer.UnmanagedObject is the type of **OdDbResbuf**;

ResultBuffer.ResbufObject is the type of **resbuf**;

Use **ResultBuffer.Create(IntPtr,bool)** to replace **ResultBuffer(IntPtr,bool)**

6.4. Alternative method to import Unmanaged ARX Function

The alternative method to import unmanaged arx function **acedCmd,acedCommand**:

Use **GrxCAD.EditorInput.Editor.Command(params object[])** to replace original functions.

6.5. Alternative Method

Autodesk.AutoCAD.EditorInput.Editor.TraceBoundary(Autodesk.AutoCAD.Geometry.Point3d, bool) alternatvie method：
GrxCAD.EditorInput.Editor.

If **TraceBoundary(Autodesk.AutoCAD.Geometry.Point3d, bool)** has not been provided，you can use the following method to replace.

```
[CommandMethod("TESTCMD")]
public void TestCommand()
{
    var db = Application.DocumentManager.MdiActiveDocument.Database;
    var ed = Application.DocumentManager.MdiActiveDocument.Editor;

    //new a DBObjectCollection for boundary set
    DBObjectCollection objColl = new DBObjectCollection();
    using (var tr = db.TransactionManager.StartTransaction())
    {
    var curSpace = tr.GetObject(db.CurrentSpaceId, OpenMode.ForRead) as
BlockTableRecord;
    var drawOrderTable = tr.GetObject(curSpace.DrawOrderTableId,
OpenMode.ForRead) as DrawOrderTable;
    ObjectIdCollection coll = drawOrderTable.GetFullDrawOrder(0);
    ObjectId entLast =    coll[coll.Count-1];
```

```
        //use "Y" or "N" whether or not to detect islands
        //instead of "0,0" when using another seed point
        ed.Command("-BOUNDARY", "A", "I", "Y", "", "0,0", "");

        coll = drawOrderTable.GetFullDrawOrder(0);
        int nIndex = coll.IndexOf(entLast);
        for (int i = nIndex; i >= 0; --i)
        {
            coll.RemoveAt(i);
        }
        //clone the boundaries created by -BOUNDARY command to DBObjectCollection
        foreach (ObjectId id in coll)
        {
            DBObject obj = tr.GetObject(id, OpenMode.ForRead) as DBObject;
            DBObject tmp = (DBObject)obj.Clone();
            objColl.Add(tmp);
        }
        //abort the transaction to avoid the boundaries created by -BOUNDARY command
posting to db
        tr.Abort();
        }

        //this is a test code to post the cloned boundaries to current space
        using (var tr = db.TransactionManager.StartTransaction())
        {
        var curSpace = tr.GetObject(db.CurrentSpaceId, OpenMode.ForWrite) as
BlockTableRecord;
        foreach (DBObject obj in objColl)
        {
            curSpace.AppendEntity((Entity)obj);
            tr.AddNewlyCreatedDBObject(obj, true);
        }
        tr.Commit();
        }

        //dispose the DBObjectCollection after used it,and dispose the item in it if
necessary,here we do
        //not dispose the DBOjbect in DBObjectCollection because we post it to current space
before
        objColl.Dispose();
}
```

6.6. Other Usages

6.6.1. Abtain GCAD Application Object under VB enveriment

After adding C:\grxsdk\inc-x86\gcax.tlb (if your GstarCAD is 64-bit, please choose the 64-bit corrosponding file) as reference, you can abtain GCAD Application object.

```
Imports GrxCAD.Runtime``
Imports GcadVbaLib

Public Class Class1
    <CommandMethod("GetGcadApplication")>
    Public Sub GetGcadApplication()

        'Define abtain GcadApplication object, the object is null
        Dim GCadApp As GcadApplication



        ' Abtain the object address through the process name.
        GCadApp = GetObject(, "gcad.application")
        'Minimize the abtained process window
        GCadApp.WindowState = GcWindowState.acMin

    End Sub
End Class
    End Sub
End Class
```

6.6.2. C#.NET and VB.NET Name space Modifaction

- VB.NET：

  The content after **Imports**，change **Autodesk.AutoCAD** to **GrxCAD**
- C#.NET：

  The content after **using**，change **Autodesk.AutoCAD** to **GrxCAD**
- If use COM：

  With VB.net, add **Imports GcadVbaLib**

  With C#.net, add **using GcadVbaLib**

  Delete **Autodesk.AutoCAD.Interop.Common**

For the COM object name in the code, such as **AcadLWPolyline**, please change the prefix Acad to Gcad, for example **AcadLWPolyline** should be change to **GcadLWPolyline**.

If some codes not sure, you can check from the object browser. For example ACAD_COLOR, you can check from the browser whether there is GCAD_COLOR, if there is, just replace it.

6.7. Replacement Sample

6.7.1. Namespace Replacement

AutoCAD	GstarCAD
Autodesk.AutoCAD.ApplicationServices	GrxCAD.ApplicationServices
Autodesk.AutoCAD.EditorInput	GrxCAD.EditorInput
Autodesk.AutoCAD.GraphicsSystem	GrxCAD.GraphicsSystem
Autodesk.AutoCAD.PlottingServices	GrxCAD.PlottingServices
Autodesk.AutoCAD.Publishing	GrxCAD.Publishing
Autodesk.AutoCAD.Runtime	GrxCAD.Runtime
Autodesk.AutoCAD.Windows	GrxCAD.Windows
Autodesk.AutoCAD.Colors	GrxCAD.Colors
Autodesk.AutoCAD.DatabaseServices	GrxCAD.DatabaseServices
Autodesk.AutoCAD.DatabaseServices.Filters	GrxCAD.DatabaseServices.Filters
Autodesk.AutoCAD.Geometry	GrxCAD.Geometry
Autodesk.AutoCAD.GraphicsInterface	GrxCAD.GraphicsInterface
Autodesk.AutoCAD.LayerManager	GrxCAD.LayerManager

6.7.2. Class Name and Property Name Replacement Sample

AutoCAD	GstarCAD
Autodesk.AutoCAD.Interop.Common.ACAD_COLOR	GcadVbaLib.GCAD_COLOR
Autodesk.AutoCAD.Interop.Common.AcadEntity	GcadVbaLib.GcadEntity
Autodesk.AutoCAD.Interop.Common.AcSelect	GcadVbaLib.GcSelect
Autodesk.AutoCAD.Interop.Common.AcadRasterImage	GcadVbaLib.GcadRasterImage
Autodesk.AutoCAD.Interop.Common.AcadText	GcadVbaLib.GcadText
Autodesk.AutoCAD.Interop.Common.AcadMText	GcadVbaLib.GcadMText
Autodesk.AutoCAD.Interop.Common.AcWindowState	GcadVbaLib.GcWindowState

7. Copyright

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